

**NATIONAL UNIVERSITY OF MODERN
LANGUAGE ISLAMABAD**

Lab 04



Submitted to:

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Submitted by:

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Roll No: SP21379

Subject: computer network

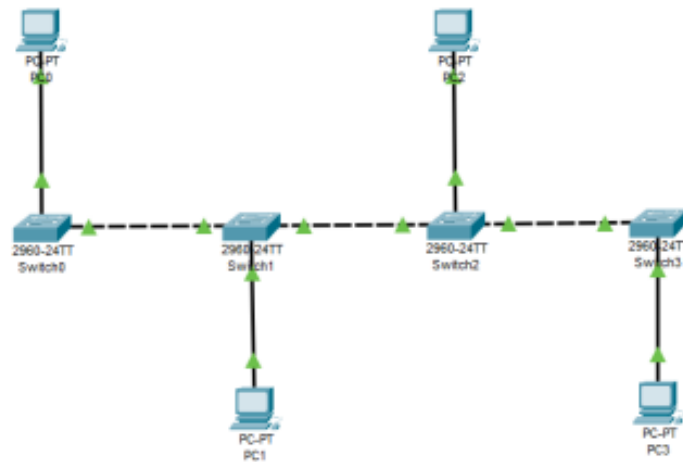
Department: BSSE-5-Afternoon

Date: 10-11-2024

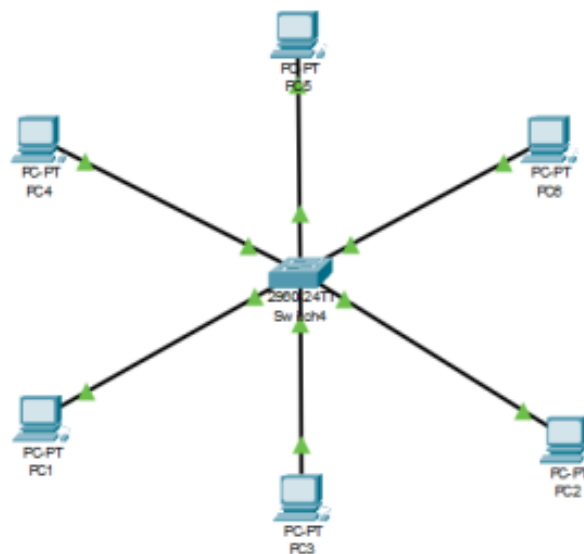
Task 1. Draw the following topologies

- a. Bus Topology
- b. Star Topology
- c. Mesh Topology.

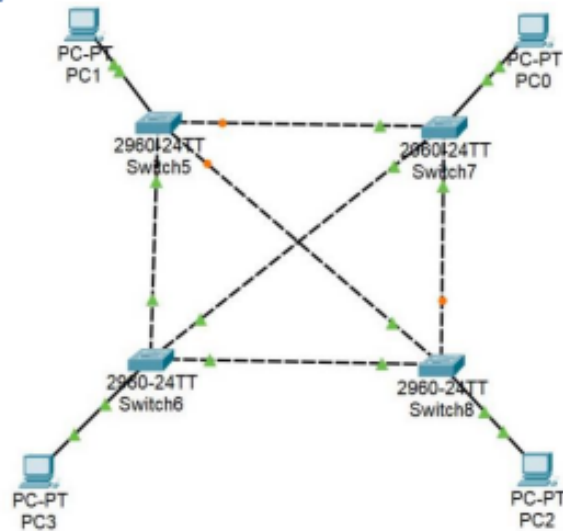
Bus Topology:



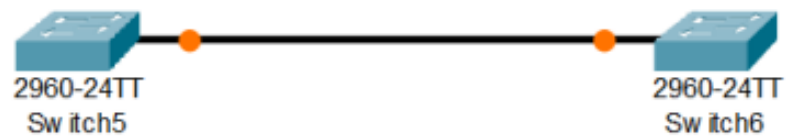
Star Topology:



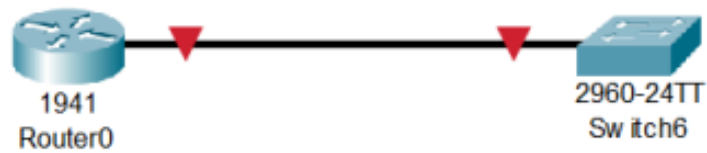
Mesh Topology:



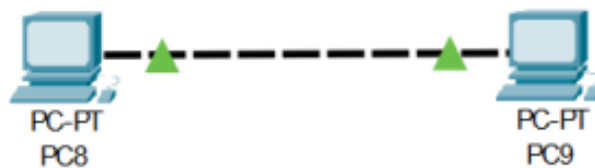
Task 2. Connect 2nd layer device to a 2nd layer devices.



Task 3. Connect 2nd layer device to a 3rd layer device.



Task 4. Connect PC to a PC



Task 5: Connect to the console



```
Terminal
Processor board ID FOC1010X104
Last reset from power-on
1 Virtual Ethernet interface
24 FastEthernet interfaces
3 Gigabit Ethernet interfaces
The password-recovery mechanism is enabled.
64K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address      : 00:60:3E:63:EB:80
Motherboard assembly number    : 73-10390-03
Power supply part number       : 341-0097-02
Motherboard serial number      : FOC10053R12
Power supply serial number     : A2S1007032M
Model revision number          : B0
Motherboard revision number    : B0
Model number                   : WS-C2960-24TT-L
System serial number           : FOC1010X104
Top Assembly Part Number       : 800-27221-02
Top Assembly Revision Number   : A0
Version ID                     : V02
CLEI Code Number               : COM3L00BRA
Hardware Board Revision Number : 0x01

Switch Ports Model          SW Version  SW Image
-----
* 1 26 WS-C2960-24TT-L 15.0(2)SE4 C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE
SOFTWARE (fcl)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnnguyen

Press RETURN to get started!

Switch>
```

Task 6: Set a management IP and default gateway

```
Switch> enable
Switch#
```

```
Switch> enable
Switch#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
```

```
Switch>enable
Switch#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vlan1
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

Switch(config-if)#
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vlan1
Switch(config-if)#ip address 192.168.1.10 255.255.255.0
Switch(config-if)#exit
Switch(config)#ip deefault-gateway 192.168.1.1
      ^
% Invalid input detected at '^' marker.

Switch(config)#ip default-gateway 192.168.1.1
Switch(config)#
```

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Task 7: Set hostname

```
Switch>
Switch>
Switch>
Switch>
Switch>en
Switch#conf
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SP21379-SWt-1
SP21379-SWt-1(config)#exit
SP21379-SWt-1#
%SYS-5-CONFIG_I: Configured from console by console

SP21379-SWt-1#
SP21379-SWt-1#
SP21379-SWt-1#
SP21379-SWt-1#
SP21379-SWt-1#
SP21379-SWt-1#
SP21379-SWt-1#
SP21379-SWt-1#
SP21379-SWt-1#
```

Task 8: Setting Duplex of Individual Port

```
SP21379-SWt-1#
SP21379-SWt-1#
SP21379-SWt-1#config
SP21379-SWt-1#configure ter
SP21379-SWt-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SP21379-SWt-1(config)#interface fas
SP21379-SWt-1(config)#interface fastEthernet 0/1
SP21379-SWt-1(config-if)#duplex full
SP21379-SWt-1(config-if)#exit
SP21379-SWt-1(config)#
SP21379-SWt-1(config)#
SP21379-SWt-1(config)#
SP21379-SWt-1(config)#
```

Task 9: Setting speed of individual port and setting up the mode

```
SP21379-SWt-1#
SP21379-SWt-1#
SP21379-SWt-1#config
SP21379-SWt-1#configure ter
SP21379-SWt-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SP21379-SWt-1(config)#interface fas
SP21379-SWt-1(config)#interface fastEthernet 0/1
SP21379-SWt-1(config-if)#duplex full
SP21379-SWt-1(config-if)#exit
SP21379-SWt-1(config)#
SP21379-SWt-1(config)#
SP21379-SWt-1(config)#
SP21379-SWt-1(config)#inter
SP21379-SWt-1(config)#interface fas
SP21379-SWt-1(config)#interface fastEthernet 0/1
SP21379-SWt-1(config-if)#speed 100
SP21379-SWt-1(config-if)#switchport mode access
SP21379-SWt-1(config-if)#
SP21379-SWt-1(config-if)#
SP21379-SWt-1(config-if)#
SP21379-SWt-1(config-if)#
```

Task 10: For multiple ports assignment (only for Switch)

```
SP21379-SWt-1(config)#
SP21379-SWt-1(config)#
SP21379-SWt-1(config)#inte
SP21379-SWt-1(config)#interface range fas
SP21379-SWt-1(config)#interface range fastEthernet 0/24
SP21379-SWt-1(config-if-range)#switch mode access
SP21379-SWt-1(config-if-range)#
SP21379-SWt-1(config-if-range)#
SP21379-SWt-1(config-if-range)#
SP21379-SWt-1(config-if-range)#
```

Task 11: Save Configuration

```
SP21379-SWt-1#
SP21379-SWt-1#
SP21379-SWt-1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
SP21379-SWt-1#
```

Task 12: Switch(config)#no ip domain lookup Router(config)#no ip domain lookup

```
SP21379-SWt-1#
SP21379-SWt-1#
SP21379-SWt-1#config
SP21379-SWt-1#configure ter
SP21379-SWt-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SP21379-SWt-1(config)#no ip domain lookup
SP21379-SWt-1(config)#exit
SP21379-SWt-1#
%SYS-5-CONFIG_I: Configured from console by console
SP21379-SWt-1#
```

Task 13: Switch(config)#line con 0 Switch(config-line)#logging synchronous Router(config)#line con 0 Router(config-line)#logging synchronous

```
SP21379-SWt-1#
SP21379-SWt-1#
SP21379-SWt-1#config
SP21379-SWt-1#configure ter
SP21379-SWt-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SP21379-SWt-1(config)#line con 0
SP21379-SWt-1(config-line)#logging sy
SP21379-SWt-1(config-line)#logging synchronous
SP21379-SWt-1(config-line)#exit
SP21379-SWt-1(config)#
```

Task 14: Show running-config

```
SP21379-SWt-1(config)#show run
      ^
% Invalid input detected at '^' marker.

SP21379-SWt-1(config)#exit
SP21379-SWt-1#
%SYS-5-CONFIG_I: Configured from console by console

SP21379-SWt-1#show run
Building configuration...

Current configuration : 1200 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname SP21379-SWt-1
!
!
!
no ip domain-lookup
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
    switchport mode access
    duplex full
    speed 100
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
--More--
```